Assignment_03_CO_01

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QUESTIONS

- **Q1** Print an X-shaped number pattern — 5 marks
- **Q2** Check whether a number is an automorphic number — 5 marks
- **Q3** Find the smallest number whose digit product equals a given value — 5 marks
- **Q4** Print Pascal's triangle — 5 marks
- **Q5** Find the next number with the same digit sum — 5 marks

5/5 answered

Q1 Print an X-shaped number pattern — 5 marks

Print an X pattern of size 5, with numbers on both diagonals.

```
1 1
2 2
3
4 4
5 5
```

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int n;
    cin >> n;

    for (int i = 1; i <= n; i++) {
        for (int j = 1; j <= n; j++) {
            if (j == i || j == n - i + 1)
                cout << i;
            else
                cout << " ";
        }
        cout << endl;
    }

    return 0;
}
```

Input

```
5
```

Output

```
1   1
 2 2
  3
 4 4
5   5
```

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5/5 answered

Q2 Check whether a number is an automorphic number — 5 marks

A number is automorphic if its square ends with the same digits as the original number. Example: \(25^2=625\), so 25 is automorphic. TC_1:25 →Automorphic number. TC_2:7→ Not an Automorphic number. TC_3:-5 → Invalid input

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    long long n;
    cin >> n;

    if (n <= 0) {
        cout << "Invalid input";
        return 0;
    }

    long long square = n * n;
    long long temp = n;
    long long power = 1;

    while (temp > 0) {
        power *= 10;
        temp /= 10;
    }

    if (square % power == n)
        cout << "Automorphic number";
    else
        cout << "Not an Automorphic number";

    return 0;
}
```

Input

```
25
```

Output

```
Automorphic number
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✓

5.00 / 5 marks

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- Q3 ✓ Find the smallest number whose digit product equals a given value — 5 marks
- Q4 ✓ Print Pascal's triangle — 5 marks
- Q5 ✓ Find the next number with the same digit sum — 5 marks

5/5 answered

Q3 Find the smallest number whose digit product equals a given value — 5 marks

Read an integer n and find the smallest positive integer whose digits multiply to n. If no such number exists, print an appropriate message.
Example: Input 36 → Output 49, because \(4 \times 9 = 36\).
TC_1:36 → Smallest number:49. TC_2:13 → No Solution. TC_3:0 → Smallest number:10

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int n, ans = 0, p = 1;
    cin >> n;

    if (n == 0) {
        cout << 10;
        return 0;
    }

    for (int i = 9; i >= 2; i--)
        while (n % i == 0) {
            ans += i * p;
            p *= 10;
            n /= i;
        }

    if (n != 1) cout << "No Solution";
    else cout << ans;

    return 0;
}
```

Input

```
36
```

Output

```
49
```

Visible Test Cases

- Test Case 1 ✗
- Test Case 2 ✓
- Test Case 3 ✗

1.67 / 5 marks

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SIMATS ENGINEERING

SIMATS Online Programming Lab

C++ PROGRAMMING

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QUESTIONS

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- **Q4** Print Pascal's triangle — 5 marks
- **Q5** Find the next number with the same digit sum — 5 marks

5/5 answered

Q4 Print Pascal's triangle — 5 marks

Print Pascal's triangle for five rows, where each number is the sum of the two numbers immediately above it.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1

Your Code

```cpp
#include <iostream>
using namespace std;

int main() {
    int n = 5;

    for (int i = 0; i < n; i++) {
        int num = 1;

        for (int j = 0; j <= i; j++) {
            cout << num << " ";
            num = num * (i - j) / (j + 1);
        }

        cout << endl;
    }

    return 0;
}
```

Input

```
stdin input
```

Output

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

Submitted

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QUESTIONS

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- Q4 Print Pascal's triangle — 5 marks
- Q5 Find the next number with the same digit sum — 5 marks

5/5 answered

Q5 Find the next number with the same digit sum — 5 marks

Read an integer and find the next greater integer having the same sum of digits. Example: Input 59 → Output 68, because both have digit sum 14. TC_1:59 → Next number:68. TC_2:123 → Next number:132. TC_3:-0→Invalid input

Your Code

```c
#include <stdio.h>
int digitSum(int n) {
    int sum = 0;

    while (n > 0) {
        sum += n % 10;
        n /= 10;
    }
    return sum;
}
int main() {
    int n, i;
    scanf("%d", &n);
    if (n < 0) {
        printf("Invalid input");
        return 0;
    }
    for (i = n + 1; ; i++) {
        if (digitSum(i) == digitSum(n)) {
            printf("Next number:%d", i);
            break;
        }
    }
    return 0;
}
```

Input

```
59
```

Output

```
Next number:68
```

Visible Test Cases

- Test Case 1 ✓
- Test Case 2 ✓
- Test Case 3 ✗

3.33 / 5 marks

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